Bi-Path Hybrid Cascade Pipeline Architecture for UWB Fall Detection

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1 Introduction and Pipeline Overview

The following diagram illustrates the comprehensive architecture of a continuous fall detection system based on Ultra-Wideband (UWB) radar signals (UWB Continuous Temporal Action Localization), integrating a deep learning pathway with a zero-phase-delay biomechanical kinematics pathway:

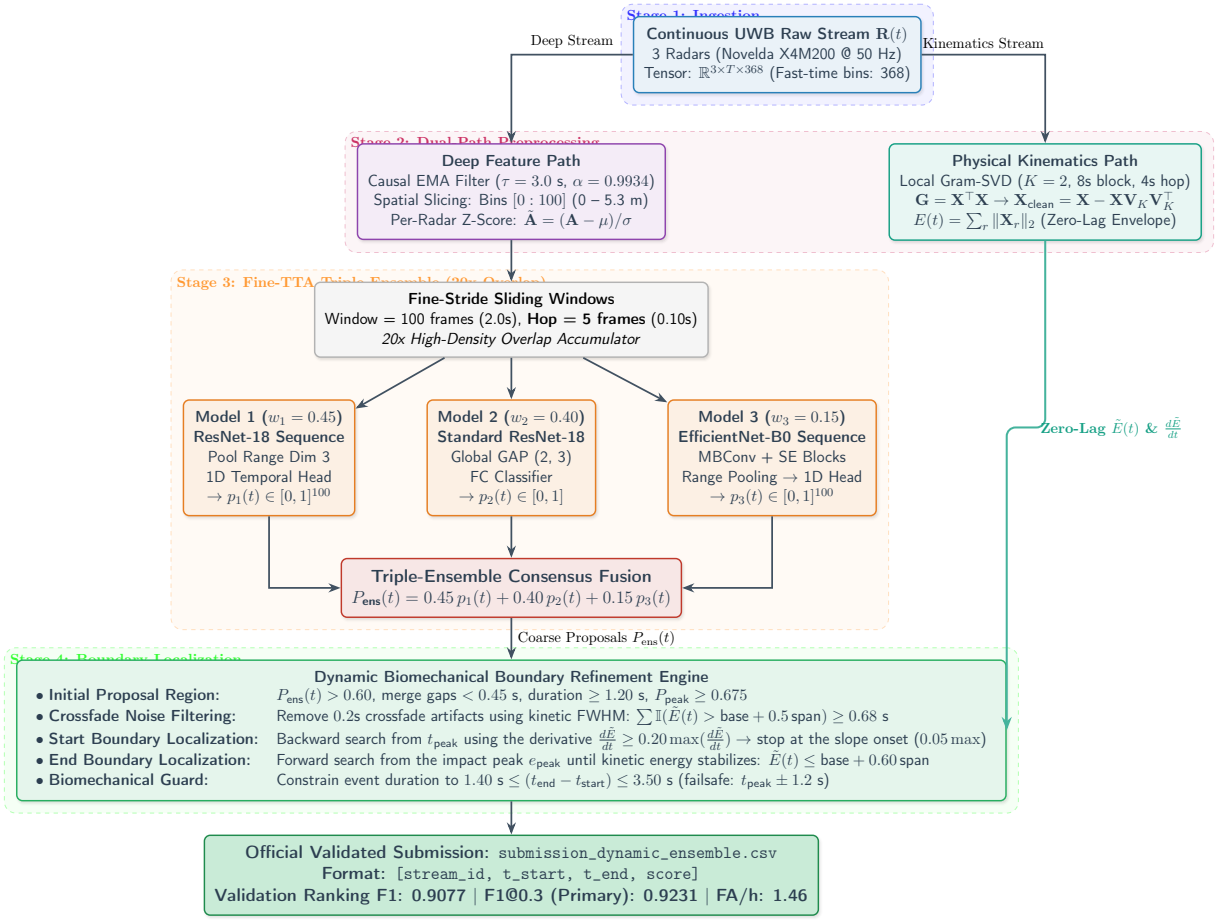


Figure 1: Overall architecture of the Bi-Path Hybrid Cascade Pipeline for continuous UWB-based fall detection.

[a4paper, 11pt]article [margin=1.8cm]geometry amsmath, amssymb graphicx tikz

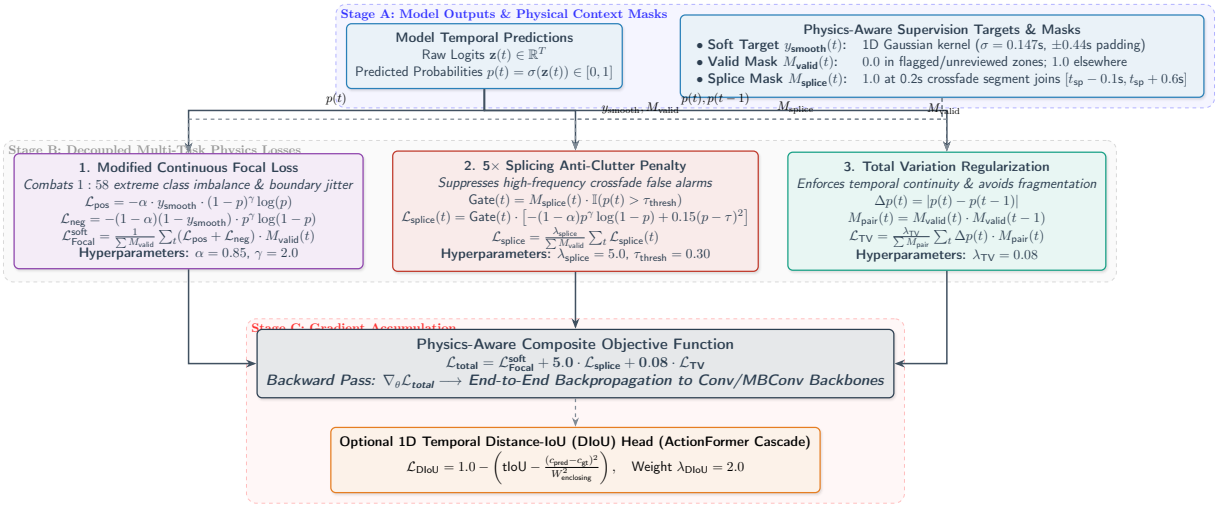


Figure 2: Comprehensive architecture of the Physics-Aware Composite Loss function, illustrating the continuous Gaussian label smoothing, the $5\times$ gated anti-splice penalty, and the Total Variation temporal regularization.